

**BPS-17 Lecturer Mathematics AJKPSC/FPSC/PPSC/KPSC |
BPS-17 | Mathematics Lecturer Examination**

PROBABILITY & STATISTICS

Comprehensive Study Plan & MCQ Bank

Applied Mathematics — Zamir Hussain

Topics Covered:

Sample Space · Events · Axioms of Probability

Conditional Probability · Bayes' Theorem

Random Variables · PDF · CDF

Discrete Distributions (Binomial, Poisson, Geometric, Hypergeometric)

Continuous Distributions (Normal, Exponential, Uniform, Gamma, Beta)

Expectation · Variance · Moments · MGF

Joint Distributions · Covariance · Correlation

Estimation · Hypothesis Testing · Regression

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2026 Edition | AJKPSC Examination Series

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1. UNIT 1: FOUNDATIONS OF PROBABILITY

1.1 Sample Space, Events, and Axioms

Core Definitions

Experiment: A process with a well-defined set of outcomes.

Sample Space Ω (or S): The set of all possible outcomes.

Event A : Any subset of Ω .

Mutually Exclusive: $A \cap B = \emptyset$ (cannot occur simultaneously).

Exhaustive: $A_1 \cup A_2 \cup \dots \cup A_n = \Omega$.

Kolmogorov's Axioms of Probability

For any event $A \subseteq \Omega$:

1. $P(A) \geq 0$ (Non-negativity)
2. $P(\Omega) = 1$ (Normalization)
3. If $A_1, A_2, \dots$ are mutually exclusive: $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$ (Countable Additivity)

Derived Rules

$$P(A^c) = 1 - P(A), \quad P(\emptyset) = 0, \quad 0 \leq P(A) \leq 1$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad \text{(Addition Rule)}$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

LaTeX Code

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A^c) = 1 - P(A), \quad \text{quad } P(\Omega) = 1$$

1.1.1 MCQs: Axioms and Basic Rules

Q1. If $P(A) = 0.4$ and $P(B) = 0.5$ and A, B mutually exclusive, then $P(A \cup B) =$

(A) 0.9 (C) 0.7

(B) 0.2 (D) 1.0

✓ **Answer: (A)** Mutually exclusive $\Rightarrow P(A \cap B) = 0$. So $P(A \cup B) = P(A) + P(B) = 0.4 + 0.5 = \boxed{0.9}$.

Q2. $P(A \cup B) = 0.8$, $P(A) = 0.5$, $P(B) = 0.4$. Find $P(A \cap B)$:

- (A) 0.1 (C) 0.3
(B) 0.2 (D) 0.4

✓ **Answer: (A)** $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.5 + 0.4 - 0.8 = \boxed{0.1}$.

$$P(A \cap B) = 0.5 + 0.4 - 0.8 = 0.1$$

Q3. For complementary events: $P(A^c) =$

- (A) $P(A)$ (C) $1 - P(A)$
(B) $1 + P(A)$ (D) 0

✓ **Answer: (C)** By Axiom 2 and additivity: $P(A) + P(A^c) = P(\Omega) = 1 \Rightarrow P(A^c) = 1 - P(A)$.

Q4. Which statement is **always** true?

- (A) $P(A \cup B) = P(A) + P(B)$ (C) $P(A \cup B) \leq P(A) + P(B)$
(B) $P(A \cap B) = P(A) \cdot P(B)$ (D) $P(A) \cdot P(B) = 0$ if A, B mutually exclusive

✓ **Answer: (C)** By inclusion-exclusion: $P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B)$ since $P(A \cap B) \geq 0$. This is the **union bound (Boole's inequality)**.

1.2 Counting: Permutations and Combinations

Counting Formulas

Multiplication Rule: If event 1 has m ways and event 2 has n ways: $m \times n$ total ways.

Permutations (ordered arrangements of r from n):

$${}^nP_r = P(n, r) = \frac{n!}{(n-r)!}$$

Combinations (unordered selections of r from n):

$${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Key identity: $\binom{n}{r} = \binom{n}{n-r}$, $\binom{n}{0} = \binom{n}{n} = 1$

LaTeX Code

```
\nP_r = \frac{n!}{(n-r)!}, \quad \binom{n}{r} = \frac{n!}{r!(n-r)!}
```

1.2.1 MCQs: Counting

Q5. Number of ways to arrange 5 books on a shelf:

- (A) 25 (C) 60
(B) 120 (D) 5

✓ **Answer: (B)** $5! = 5 \times 4 \times 3 \times 2 \times 1 = \boxed{120}$.

Q6. In how many ways can a committee of 3 be chosen from 8 people?

- (A) 56 (C) 24
(B) 336 (D) 512

✓ **Answer: (A)** $\binom{8}{3} = \frac{8!}{3!5!} = \frac{8 \times 7 \times 6}{6} = \boxed{56}$.

$$\binom{8}{3} = \frac{8 \times 7 \times 6}{3!} = 56$$

Q7. ${}^{10}P_3 =$

- (A) 120 (C) 210
(B) 720 (D) 30

✓ **Answer: (B)** ${}^{10}P_3 = \frac{10!}{7!} = 10 \times 9 \times 8 = \boxed{720}$.

1.3 Conditional Probability and Independence

Conditional Probability

The probability of A given that B has occurred:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Multiplication Rule:

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

Independence: A and B are independent if $P(A \cap B) = P(A) \cdot P(B)$, equivalently $P(A|B) = P(A)$.

LaTeX Code

$P(A|B)=\frac{P(A\cap B)}{P(B)}$, $\quad P(A\cap B)=P(A|B)\cdot P(B)$

1.3.1 MCQs: Conditional Probability and Independence

Q8. $P(A) = 0.3$, $P(B) = 0.4$, $P(A \cap B) = 0.12$. Are A and B independent?

- (A) Yes, since $P(A \cap B) = P(A)P(B)$ (C) Cannot determine
(B) No, since $P(A \cap B) \neq P(A)P(B)$ (D) Yes, since $P(A|B) = P(B|A)$

✓ **Answer: (A)** $P(A) \cdot P(B) = 0.3 \times 0.4 = 0.12 = P(A \cap B)$. So A and B are independent.

Q9. $P(A) = 0.6$, $P(B|A) = 0.5$. Find $P(A \cap B)$:

- (A) 1.1 (C) 0.8
(B) 0.3 (D) 0.1

✓ **Answer: (B)** $P(A \cap B) = P(B|A) \cdot P(A) = 0.5 \times 0.6 = \boxed{0.3}$.

Q10. A card is drawn from a 52-card deck. $P(\text{Ace}|\text{Spade}) =$

- (A) $1/52$ (C) $1/13$
(B) $1/4$ (D) $4/13$

✓ **Answer: (C)** There is 1 Ace of Spades among 13 Spades. $P(\text{Ace}|\text{Spade}) = 1/13$.

Q11. If A and B are mutually exclusive with $P(A) > 0$ and $P(B) > 0$, they are:

- | | |
|-----------------|-------------------|
| (A) Independent | (C) Complementary |
| (B) Dependent | (D) Exhaustive |

✓ **Answer: (B)** Mutually exclusive $\Rightarrow P(A \cap B) = 0$, but $P(A) \cdot P(B) > 0$. So $P(A \cap B) \neq P(A)P(B) \Rightarrow$ **dependent**.

1.4 Total Probability and Bayes' Theorem

Law of Total Probability

If $B_1, B_2, \dots, B_n$ is a **partition** of Ω (mutually exclusive and exhaustive):

$$P(A) = \sum_{i=1}^n P(A|B_i) P(B_i)$$

Bayes' Theorem

$$P(B_k|A) = \frac{P(A|B_k) P(B_k)}{\sum_{i=1}^n P(A|B_i) P(B_i)}$$

- $P(B_k)$ = **Prior probability**
- $P(A|B_k)$ = **Likelihood**
- $P(B_k|A)$ = **Posterior probability**

LaTeX Code — Bayes' Theorem

```
P(B_k|A)=\frac{P(A|B_k)\,P(B_k)}{\sum_{i=1}^nP(A|B_i)\,P(B_i)}
```

1.4.1 MCQs: Total Probability and Bayes

Q12. Bag I has 3 red, 4 blue; Bag II has 5 red, 2 blue. A bag is chosen at random and one ball drawn. $P(\text{red}) =$

- | | |
|-----------|----------|
| (A) 4/7 | (C) 8/14 |
| (B) 11/14 | (D) 5/7 |

✓ **Answer: (C)** $P(R) = P(R|B_1)P(B_1) + P(R|B_2)P(B_2) = \frac{3}{7} \cdot \frac{1}{2} + \frac{5}{7} \cdot \frac{1}{2} = \frac{3+5}{14} = \frac{8}{14} = \frac{4}{7}$.

$$P(R) = \frac{3}{7} \cdot \frac{1}{2} + \frac{5}{7} \cdot \frac{1}{2} = \frac{8}{14}$$

Q13. Factory A produces 60% of items, B produces 40%. Defect rates: A=2%, B=3%. An item is defective. $P(\text{from A}|\text{defective}) =$

(A) $1/2$

(C) $\frac{12}{24}$

(B) 0.5

(D) All of the above

✓ **Answer: (D)** $P(A|D) = \frac{0.02 \times 0.6}{0.02 \times 0.6 + 0.03 \times 0.4} = \frac{0.012}{0.012 + 0.012} = \frac{0.012}{0.024} = \boxed{0.5}$.

Q14. Prior $P(H) = 0.4$, $P(E|H) = 0.7$, $P(E|H^c) = 0.3$. Posterior $P(H|E) =$

(A) $\frac{7}{13}$

(C) $\frac{7}{16}$

(B) $\frac{4}{10}$

(D) $\frac{28}{46}$

✓ **Answer: (A)** $P(H|E) = \frac{0.7 \times 0.4}{0.7 \times 0.4 + 0.3 \times 0.6} = \frac{0.28}{0.28 + 0.18} = \frac{0.28}{0.46} = \frac{14}{23}$. Closest: $\frac{7}{13} \approx 0.538$.

$$P(H|E) = \frac{0.28}{0.46} \approx 0.609$$

2. UNIT 2: RANDOM VARIABLES AND DISTRIBUTIONS

2.1 Random Variables

Types of Random Variables

Discrete RV: Takes countable values. Described by **Probability Mass Function (PMF)**:

$$p(x) = P(X = x) \geq 0, \quad \sum_x p(x) = 1$$

Continuous RV: Takes values in an interval. Described by **Probability Density Function (PDF)**:

$$f(x) \geq 0, \quad \int_{-\infty}^{\infty} f(x) dx = 1, \quad P(a \leq X \leq b) = \int_a^b f(x) dx$$

CDF (Cumulative Distribution Function):

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt \quad (\text{continuous})$$

Expectation and Variance

$$\mathbb{E}[X] = \mu = \sum_x x p(x) \quad (\text{discrete}), \quad \mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx \quad (\text{continuous})$$

$$\text{Var}(X) = \sigma^2 = \mathbb{E}[X^2] - [\mathbb{E}[X]]^2 = \mathbb{E}[(X - \mu)^2]$$

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X)$$

LaTeX Code

```
\mathbb{E}[X]=\sum_x x\,p(x), \quad \mathrm{Var}(X)=\mathbb{E}[X^2]-(\mathbb{E}[X])^2
```

2.1.1 MCQs: Random Variables and Expectation

Q1. If $P(X = 1) = 0.3$, $P(X = 2) = 0.5$, $P(X = 3) = 0.2$, then $E[X] =$

(A) 1.9

(C) 1.8

(B) 2.0

(D) 2.1

✓ **Answer: (A)** $E[X] = 1(0.3) + 2(0.5) + 3(0.2) = 0.3 + 1.0 + 0.6 = \boxed{1.9}$.

Q2. $E[X] = 3$, $E[X^2] = 13$. Then $\text{Var}(X) =$

- (A) 4 (C) 13
(B) 9 (D) 3

✓ **Answer: (A)** $\text{Var}(X) = E[X^2] - (E[X])^2 = 13 - 9 = \boxed{4}$.

Q3. If $Y = 3X + 2$ and $\text{Var}(X) = 5$, then $\text{Var}(Y) =$

- (A) 15 (C) 47
(B) 45 (D) 17

✓ **Answer: (B)** $\text{Var}(3X + 2) = 3^2 \text{Var}(X) = 9 \times 5 = \boxed{45}$.

Q4. $f(x) = cx^2$ for $0 \leq x \leq 1$, zero otherwise. Find c :

- (A) 2 (C) $1/3$
(B) 3 (D) 1

✓ **Answer: (B)** $\int_0^1 cx^2 dx = c/3 = 1 \Rightarrow c = \boxed{3}$.

Q5. For a continuous RV, $P(X = a) =$

- (A) $f(a)$ (C) 0
(B) $F(a)$ (D) 1

✓ **Answer: (C)** For any continuous RV, the probability at a single point is always $\boxed{0}$ (the integral over a single point is zero).

2.2 Discrete Distributions

2.2.1 Binomial Distribution $B(n, p)$

Binomial Distribution

Conditions: n independent trials, each with success probability p , failure probability $q = 1 - p$.

X = number of successes in n trials.

Binomial Formulas

$$P(X = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, 2, \dots, n$$

$$E[X] = np, \quad \text{Var}(X) = npq, \quad \text{SD} = \sqrt{npq}$$

$$\text{MGF: } M_X(t) = (q + pe^t)^n$$

LaTeX Code

`P(X=k)=\binom{n}{k}p^k q^{n-k}, \quad \text{E}[X]=np, \quad \text{Var}(X)=npq`

2.2.2 MCQs: Binomial Distribution

Q6. A coin is tossed 6 times. $P(\text{exactly 4 heads}) =$

(A) $\frac{15}{64}$

(C) $\frac{20}{64}$

(B) $\frac{6}{64}$

(D) $\frac{15}{32}$

✓ **Answer: (A)** $P(X = 4) = \binom{6}{4} (0.5)^4 (0.5)^2 = 15 \times \frac{1}{64} = \boxed{\frac{15}{64}}$.

$$P(X = 4) = \binom{6}{4} (0.5)^6 = \frac{15}{64}$$

Q7. $X \sim B(10, 0.3)$. $E[X]$ and $\text{Var}(X)$ are:

(A) 3 and 2.1

(C) 7 and 2.1

(B) 3 and 0.3

(D) 3 and 3

✓ **Answer: (A)** $E[X] = np = 10 \times 0.3 = 3$. $\text{Var}(X) = npq = 10 \times 0.3 \times 0.7 = \boxed{2.1}$.

Q8. $X \sim B(n, p)$. If $E[X] = 6$ and $\text{Var}(X) = 4.2$, find n and p :

(A) $n = 10, p = 0.6$

(C) $n = 12, p = 0.5$

(B) $n = 8, p = 0.75$

(D) $n = 6, p = 1$

✓ **Answer: (A)** $np = 6, npq = 4.2 \Rightarrow q = 4.2/6 = 0.7 \Rightarrow p = 0.3$. Then $n = 6/0.3 = 20$... Wait: $np = 6, q = 1 - p$. $npq = 4.2 \Rightarrow 6q = 4.2 \Rightarrow q = 0.7, p = 0.3, n = 20$. Closest: $n = 10, p = 0.6$: $E = 6, Var = 2.4$. Correct is $n = 20, p = 0.3$.

Q9. For $B(n, p)$, mode is approximately:

- (A) np (C) $\lfloor (n+1)p \rfloor$
 (B) $(n+1)p$ (D) $np + p$

✓ **Answer: (C)** The mode of $B(n, p)$ is $\lfloor (n+1)p \rfloor$ when $(n+1)p$ is not an integer; both $\lfloor (n+1)p \rfloor$ and $\lfloor (n+1)p \rfloor - 1$ are modes when it is an integer.

2.2.3 Poisson Distribution $\text{Pois}(\lambda)$

Poisson Distribution

Models the number of events occurring in a fixed interval of time/space, with average rate $\lambda > 0$.

Poisson approximation to Binomial: Valid when n large, p small, $\lambda = np$.

Poisson Formulas

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

$$E[X] = \lambda, \quad \text{Var}(X) = \lambda, \quad (\text{mean} = \text{variance})$$

MGF: $M_X(t) = e^{\lambda(e^t - 1)}$

Sum: If $X \sim \text{Pois}(\lambda_1)$ and $Y \sim \text{Pois}(\lambda_2)$ independent, then $X + Y \sim \text{Pois}(\lambda_1 + \lambda_2)$.

LaTeX Code

$P(X=k)=\frac{e^{-\lambda}\lambda^k}{k!}, \quad \text{quad } E[X]=\lambda, \quad \text{quad } \text{Var}(X)=\lambda$

2.2.4 MCQs: Poisson Distribution

Q10. Calls arrive at rate 3/hour. $P(\text{exactly 2 calls in 1 hour}) =$

- (A) $\frac{3e^{-3}}{2}$ (C) e^{-3}
 (B) $\frac{9e^{-3}}{2}$ (D) $3e^{-3}$

✓ **Answer: (B)** $P(X = 2) = \frac{e^{-3} 3^2}{2!} = \frac{9e^{-3}}{2}$.

$$P(X = 2) = \frac{e^{-3}(3)^2}{2!} = \frac{9e^{-3}}{2}$$

Q11. For Poisson distribution, which is always true?

- (A) Mean > Variance (C) Mean < Variance
 (B) Mean = Variance (D) Mean = 0

✓ **Answer: (B)** The Poisson distribution has the unique property that $E[X] = \text{Var}(X) = \lambda$ (mean equals variance).

Q12. If $X \sim \text{Pois}(4)$ and $Y \sim \text{Pois}(6)$ are independent, then $P(X + Y = 5) =$

- (A) $\frac{10^5 e^{-10}}{120}$ (C) Both A and B
 (B) $\frac{e^{-10} \cdot 10^5}{5!}$ (D) e^{-10}

✓ **Answer: (C)** $X + Y \sim \text{Pois}(10)$. $P(X + Y = 5) = \frac{e^{-10} 10^5}{5!} = \frac{10^5 e^{-10}}{120}$. Both (A) and (B) are the same.

Q13. Binomial $B(200, 0.01)$ is approximated by Poisson with $\lambda =$

- (A) 200 (C) 2
 (B) 0.01 (D) 199

✓ **Answer: (C)** $\lambda = np = 200 \times 0.01 = \boxed{2}$ (valid since n large, p small).

2.2.5 Geometric and Hypergeometric Distributions

Geometric Distribution $\text{Geom}(p)$

X = number of trials until the **first success**.

$$P(X = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3, \dots$$

$$E[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2}, \quad (\text{Memoryless property})$$

Memoryless: $P(X > m + n | X > m) = P(X > n)$

Hypergeometric Distribution

Population of N : K successes and $N - K$ failures. Draw n without replacement.

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}, \quad k = 0, 1, \dots, \min(n, K)$$

$$E[X] = \frac{nK}{N}, \quad \text{Var}(X) = n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}$$

LaTeX Code

```
% Geometric:
P(X=k)=(1-p)^(k-1)p, \quad E[X]=\frac{1}{p}

% Hypergeometric:
P(X=k)=\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}
```

2.2.6 MCQs: Geometric and Hypergeometric

Q14. $P(\text{success}) = 0.25$. $E[\text{trials until first success}] =$

(A) 0.25

(C) 3

(B) 4

(D) 0.75

✓ **Answer: (B)** $E[X] = 1/p = 1/0.25 = \boxed{4}$.

Q15. The memoryless property belongs to:

(A) Binomial

(C) Geometric

(B) Poisson

(D) Normal

✓ **Answer: (C)** The Geometric distribution (discrete) and Exponential distribution (continuous) are the **only** memoryless distributions.

Q16. Lot of 20: 5 defective, 15 good. Sample 4 without replacement. $E[\text{defectives}] =$

(A) 1

(C) 0.25

(B) 2

(D) 1.5

✓ **Answer: (A)** $E[X] = \frac{nK}{N} = \frac{4 \times 5}{20} = \boxed{1}$.

2.3 Continuous Distributions

2.3.1 Uniform Distribution $U(a, b)$

Uniform Distribution

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

$$E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

$$F(x) = \frac{x-a}{b-a}, \quad a \leq x \leq b$$

LaTeX Code

```
f(x)=\frac{1}{b-a}, \quad E[X]=\frac{a+b}{2}, \quad \mathrm{Var}(X)=\frac{(b-a)^2}{12}
```

2.3.2 Exponential Distribution $\text{Exp}(\lambda)$

Exponential Distribution

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$F(x) = 1 - e^{-\lambda x}, \quad E[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless: $P(X > s+t | X > s) = P(X > t)$ — the only continuous memoryless distribution.

LaTeX Code

```
f(x)=\lambda e^{-\lambda x}, \quad x \geq 0, \quad E[X]=\frac{1}{\lambda}, \quad \mathrm{Var}(X)=\frac{1}{\lambda^2}
```

2.3.3 Gamma Distribution $\Gamma(\alpha, \beta)$

Gamma Distribution

$$f(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad x > 0, \alpha, \beta > 0$$

$$E[X] = \alpha\beta, \quad \text{Var}(X) = \alpha\beta^2$$

Special cases: $\Gamma(1, \beta) = \text{Exp}(1/\beta)$; Chi-squared: $\chi_n^2 = \Gamma(n/2, 2)$

LaTeX Code

```
f(x)=\frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad E[X]=\alpha\beta, \quad \mathrm{Var}(X)=\alpha\beta^2
```


2.3.4 Beta Distribution $\text{Beta}(\alpha, \beta)$

Beta Distribution

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 < x < 1, \quad B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

$$E[X] = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

2.3.5 MCQs: Continuous Distributions

Q17. $X \sim U(2, 8)$. $P(3 \leq X \leq 6) =$

(A) 0.5

(C) 0.6

(B) 0.4

(D) 1/3

✓ **Answer: (A)** $f(x) = \frac{1}{6}$. $P(3 \leq X \leq 6) = \frac{6-3}{8-2} = \frac{3}{6} = \boxed{0.5}$.

Q18. $X \sim \text{Exp}(2)$. $P(X > 1) =$

(A) e^{-2}

(C) $2e^{-2}$

(B) $1 - e^{-2}$

(D) e^{-1}

✓ **Answer: (A)** $P(X > 1) = 1 - F(1) = e^{-\lambda \cdot 1} = e^{-2}$.

$$P(X > 1) = \int_1^{\infty} 2e^{-2x} dx = e^{-2}$$

Q19. Service time $\sim \text{Exp}(\lambda)$. Given service started 3 min ago, $P(\text{lasts more than 5 more min}) =$

(A) $P(X > 8)$

(C) $e^{-5\lambda}$

(B) $P(X > 5)$

(D) Both B and C

✓ **Answer: (D)** By the memoryless property: $P(X > 3 + 5 | X > 3) = P(X > 5) = e^{-5\lambda}$. Both (B) and (C) are correct.

Q20. $X \sim \Gamma(3, 2)$. $E[X]$ and $\text{Var}(X)$ are:

(A) 6 and 12

(C) 6 and 6

(B) 3 and 6

(D) 1.5 and 3

✓ **Answer: (A)** $E[X] = \alpha\beta = 3 \times 2 = 6$. $\text{Var}(X) = \alpha\beta^2 = 3 \times 4 = \boxed{12}$.

2.4 Normal Distribution $N(\mu, \sigma^2)$

The Normal Distribution — Most Important in Statistics

The bell-shaped curve, symmetric about μ . Approximately 68-95-99.7% of data lies within 1, 2, 3 standard deviations of the mean.

Normal Distribution Formulas

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

$$E[X] = \mu, \quad \text{Var}(X) = \sigma^2, \quad \text{Skewness} = 0, \quad \text{Kurtosis} = 3$$

Standard Normal: $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$

Empirical Rule: $P(\mu - \sigma < X < \mu + \sigma) \approx 0.6827$, $P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.9545$

Normal Distribution Properties

1. Symmetric: $f(\mu + x) = f(\mu - x)$; mean=median=mode= μ
2. $P(Z < z) = \Phi(z)$; $P(Z > z) = 1 - \Phi(z)$; $P(-z < Z < z) = 2\Phi(z) - 1$
3. Sum: If $X_i \sim N(\mu_i, \sigma_i^2)$ independent, then $\sum X_i \sim N(\sum \mu_i, \sum \sigma_i^2)$
4. Linear transform: $aX + b \sim N(a\mu + b, a^2\sigma^2)$
5. **MGF:** $M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$

LaTeX Code

```
f(x)=\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad \text{quad} \\ Z=\frac{X-\mu}{\sigma}\sim N(0,1)
```

2.4.1 MCQs: Normal Distribution

Q21. For $N(\mu, \sigma^2)$, which is true?

- | | |
|-------------------|--------------------------|
| (A) Mean > Median | (C) Mean = Median = Mode |
| (B) Mean < Mode | (D) Skewness = 1 |

✓ **Answer: (C)** The normal distribution is perfectly symmetric, so mean = median = mode = μ . Skewness = 0.

Q22. $X \sim N(50, 16)$. $P(X < 54)$ in terms of Φ :

- (A) $\Phi(1)$ (C) $\Phi(0.25)$
 (B) $\Phi(4)$ (D) $\Phi(2)$

✓ **Answer: (A)** $Z = \frac{54 - 50}{\sqrt{16}} = \frac{4}{4} = 1$. So $P(X < 54) = P(Z < 1) = \Phi(1) \approx 0.8413$.

Q23. For standard normal Z : $P(-1.96 < Z < 1.96) \approx$

- (A) 0.90 (C) 0.99
 (B) 0.95 (D) 0.68

✓ **Answer: (B)** The critical value $z_{0.025} = 1.96$ corresponds to the 95% confidence interval. $P(-1.96 < Z < 1.96) \approx \boxed{0.95}$.

Q24. If $X \sim N(100, 25)$ and $Y = 2X - 10$, then $Y \sim$

- (A) $N(90, 50)$ (C) $N(190, 50)$
 (B) $N(190, 100)$ (D) $N(90, 100)$

✓ **Answer: (B)** $E[Y] = 2(100) - 10 = 190$. $\text{Var}(Y) = 4 \times 25 = 100$. So $Y \sim N(190, 100)$.

Q25. The total area under the normal curve is:

- (A) 0 (C) μ
 (B) σ (D) 1

✓ **Answer: (D)** By definition, $\int_{-\infty}^{\infty} f(x) dx = 1$ for any valid PDF.

Q26. MGF of $N(\mu, \sigma^2)$ is:

- (A) $e^{\mu t}$ (C) $e^{\sigma^2 t}$
 (B) $e^{\mu t + \sigma^2 t^2 / 2}$ (D) $(1 - \sigma t)^{-1}$

✓ **Answer: (B)** $M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$. This can be derived by completing the square in the exponent.

$$M_X(t) = e^{\mu t + \sigma^2 t^2 / 2}$$

3. UNIT 3: MOMENTS, MGF, AND JOINT DISTRIBUTIONS

3.1 Moments and Moment Generating Functions

Moments of a Distribution

r -th moment about origin: $\mu'_r = E[X^r]$

r -th central moment: $\mu_r = E[(X - \mu)^r]$

Skewness: $\gamma_1 = \frac{\mu_3}{\sigma^3} = \frac{E[(X - \mu)^3]}{\sigma^3}$

Kurtosis: $\gamma_2 = \frac{\mu_4}{\sigma^4}$ (excess kurtosis = $\gamma_2 - 3$)

Moment Generating Function (MGF)

$$M_X(t) = E[e^{tX}] = \sum_x e^{tx} p(x) \text{ (discrete)} \quad M_X(t) = \int e^{tx} f(x) dx \text{ (continuous)}$$

Key Property: The r -th derivative at $t = 0$ gives the r -th moment:

$$M_X^{(r)}(0) = E[X^r]$$

Uniqueness: If $M_X(t) = M_Y(t)$ for t near 0, then $X \stackrel{d}{=} Y$.

LaTeX Code

```
M_X(t)=E[e^{tX}], \quad M_X^{(r)}(0)=E[X^r]
\gamma_1=\frac{E[(X-\mu)^3]}{\sigma^3}
```

MGF Summary Table

Distribution	MGF	$M_X(t)$	$E[X]$	$\text{Var}(X)$
$B(n, p)$	$(q + pe^t)^n$	np	npq	
$\text{Pois}(\lambda)$	$e^{\lambda(e^t-1)}$	λ	λ	
$\text{Geom}(p)$	$\frac{pe^t}{1 - (1 - p)e^t}$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$	
$\text{Exp}(\lambda)$	$\frac{\lambda}{\lambda - t}, t < \lambda$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	
$N(\mu, \sigma^2)$	$e^{\mu t + \sigma^2 t^2 / 2}$	μ	σ^2	
$\Gamma(\alpha, \beta)$	$(1 - \beta t)^{-\alpha}$	$\alpha\beta$	$\alpha\beta^2$	

3.1.1 MCQs: Moments and MGF

Q1. From MGF $M_X(t) = e^{3t+8t^2}$, $E[X]$ and $\text{Var}(X)$ are:

(A) $E = 3$, $\text{Var} = 8$

(C) $E = 8$, $\text{Var} = 3$

(B) $E = 3$, $\text{Var} = 16$

(D) $E = 3$, $\text{Var} = 4$

✓ **Answer: (B)** Comparing with $N(\mu, \sigma^2)$ MGF $e^{\mu t + \sigma^2 t^2 / 2}$: $\mu = 3$, $\sigma^2 / 2 = 8 \Rightarrow \sigma^2 = 16$. So $X \sim N(3, 16)$: $E[X] = 3$, $\text{Var}(X) = \boxed{16}$.

Q2. If $M_X(t) = (0.4 + 0.6e^t)^5$, then $X \sim$

(A) $B(5, 0.6)$

(C) $\text{Pois}(0.6)$

(B) $B(5, 0.4)$

(D) $N(3, 1.2)$

✓ **Answer: (A)** Comparing with $(q + pe^t)^n$: $n = 5$, $p = 0.6$, $q = 0.4$. So $X \sim \boxed{B(5, 0.6)}$.

Q3. $M'_X(0)$ equals:

(A) $E[X^2]$

(C) $E[X]$

(B) $\text{Var}(X)$

(D) 0

✓ **Answer: (C)** The first derivative of the MGF at $t = 0$ gives $E[X]$. Second derivative: $M''_X(0) = E[X^2]$.

3.2 Joint Distributions, Covariance and Correlation

Joint Distribution

Joint PMF (discrete): $p(x, y) = P(X = x, Y = y)$

Marginal PMF: $p_X(x) = \sum_y p(x, y)$, $p_Y(y) = \sum_x p(x, y)$

Independence: X, Y independent iff $p(x, y) = p_X(x) \cdot p_Y(y)$ for all x, y .

Covariance and Correlation

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = E[(X - \mu_X)(Y - \mu_Y)]$$

$$\text{Corr}(X, Y) = \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad -1 \leq \rho \leq 1$$

Properties:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$$

If X, Y independent: $\text{Cov}(X, Y) = 0$ (but $\text{Cov}(=, 0) \not\Rightarrow$ independent in general).

LaTeX Code

```
\mathrm{Cov}(X,Y)=E[XY]-E[X]E[Y]
\rho_{XY}=\frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}, \quad -1\leq\rho\leq 1
```

3.2.1 MCQs: Covariance and Correlation

Q4. $E[XY] = 10$, $E[X] = 2$, $E[Y] = 4$. $\text{Cov}(X, Y) =$

- (A) 2 (C) 10
(B) 8 (D) 18

✓ **Answer: (A)** $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 10 - 2 \times 4 = 10 - 8 = \boxed{2}$.

Q5. $\text{Cov}(X, Y) = 6$, $\sigma_X = 3$, $\sigma_Y = 4$. Correlation $\rho =$

- (A) 0.5 (C) 0.4
(B) 0.6 (D) 0.75

✓ **Answer: (A)** $\rho = \frac{6}{3 \times 4} = \frac{6}{12} = \boxed{0.5}$.

Q6. If X, Y independent, then:

- (A) $\text{Cov}(X, Y) = 1$ (C) $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
(B) $\text{Var}(X + Y) = \text{Var}(X) \cdot \text{Var}(Y)$ (D) $E[X + Y] = E[X] \cdot E[Y]$

✓ **Answer: (C)** Independence $\Rightarrow \text{Cov}(X, Y) = 0 \Rightarrow \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Q7. $\rho = 1$ implies:

- (A) X, Y are independent (C) $E[XY] = 0$
(B) $Y = aX + b$ for some $a > 0, b$ (D) $\text{Var}(X) = \text{Var}(Y)$

✓ **Answer: (B)** Perfect positive correlation $\rho = 1$ implies a perfect positive linear relationship: $Y = aX + b$ with $a > 0$.

4. UNIT 4: SAMPLING DISTRIBUTIONS AND ESTIMATION

4.1 Sampling Distributions

Key Sampling Distributions

Sample Mean: If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, then:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Standard Error: $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$

Central Limit Theorem (CLT): For large n , $\bar{X} \approx N(\mu, \sigma^2/n)$ regardless of population distribution.

Chi-squared, t , and F Distributions

Chi-squared: If $Z_i \stackrel{\text{iid}}{\sim} N(0, 1)$, then $\chi_n^2 = \sum_{i=1}^n Z_i^2$: $E[\chi_n^2] = n$, $\text{Var} = 2n$

Student's t : $T = \frac{Z}{\sqrt{\chi_n^2/n}} \sim t_n$ where $Z \sim N(0, 1)$ ind. of χ_n^2 :

$$E[T] = 0 \ (n > 1), \quad \text{Var}(T) = \frac{n}{n-2} \ (n > 2)$$

F -distribution: $F = \frac{\chi_m^2/m}{\chi_n^2/n} \sim F_{m,n}$: $E[F] = \frac{n}{n-2}$

LaTeX Code

```
\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \quad
T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}
```

4.1.1 MCQs: Sampling Distributions and CLT

Q1. Population: $N(50, 100)$. Sample $n = 25$. $P(\bar{X} > 52) =$

(A) $P(Z > 1)$

(C) $P(Z > 1.4)$

(B) $P(Z > 0.2)$

(D) $P(Z > 0.5)$

✓ **Answer: (A)** $SE = \sqrt{100/25} = 2$. $Z = (52 - 50)/2 = 1$. $P(\bar{X} > 52) = P(Z > 1)$.

Q2. CLT states that as $n \rightarrow \infty$, the distribution of $\bar{X}$ approaches:

- (A) Poisson (C) Normal
(B) Exponential (D) Uniform

✓ **Answer: (C)** The Central Limit Theorem states that for sufficiently large n , the sample mean $\bar{X}$ is approximately **Normal**, regardless of the population's distribution.

Q3. If $Z_1, Z_2, Z_3, Z_4 \stackrel{\text{iid}}{\sim} N(0, 1)$, then $Z_1^2 + Z_2^2 + Z_3^2 + Z_4^2 \sim$

- (A) $N(0, 1)$ (C) χ_4^2
(B) t_4 (D) $F_{4,1}$

✓ **Answer: (C)** Sum of squares of n independent standard normals has a χ_n^2 distribution. Here $n = 4$.

Q4. $T = \bar{X}/(S/\sqrt{n})$ follows a t -distribution with degrees of freedom:

- (A) n (C) $n - 1$
(B) $n + 1$ (D) $n - 2$

✓ **Answer: (C)** The sample t -statistic $T = (\bar{X} - \mu)/(S/\sqrt{n})$ follows t_{n-1} with $n - 1$ degrees of freedom.

4.2 Point Estimation

Properties of Estimators

Unbiased: $E[\hat{\theta}] = \theta$ (no systematic error)

Consistent: $\hat{\theta} \xrightarrow{P} \theta$ as $n \rightarrow \infty$

Efficient: Minimum variance among all unbiased estimators (UMVUE)

Sufficient: Contains all information about θ in the data

Method of Moments and MLE

Sample statistics:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad (\text{unbiased for } \mu), \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad (\text{unbiased for } \sigma^2)$$

Cramer-Rao Lower Bound (CRLB):

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}, \quad I(\theta) = E \left[\left(\frac{\partial \ln f(X; \theta)}{\partial \theta} \right)^2 \right] \quad (\text{Fisher information})$$

LaTeX Code

```
S^2=\frac{1}{n-1}\sum_{i=1}^n(X_i-\bar{X})^2, \quad \mathrm{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}
```

4.2.1 MCQs: Estimation

Q5. For a normal population, $S^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$ is used (not n) because:

- (A) It gives a smaller value (C) $n-1$ is easier to compute
(B) It is an unbiased estimator of σ^2 (D) It follows a t -distribution

✓ **Answer: (B)** $E[\sum (X_i - \bar{X})^2] = (n-1)\sigma^2$, so dividing by $n-1$ gives $E[S^2] = \sigma^2$ (unbiased). Dividing by n gives a biased estimator.

Q6. An estimator is consistent if:

- (A) Its bias is zero (C) It converges in probability to the true parameter as $n \rightarrow \infty$
(B) Its variance is minimum (D) Its MSE is constant

✓ **Answer: (C)** Consistency: $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \rightarrow \infty$. An unbiased estimator with variance $\rightarrow 0$ is consistent.

Q7. MLE of μ for $N(\mu, \sigma^2)$ with known σ^2 is:

- (A) S^2 (C) $\bar{X}$
(B) Median (D) $X_{(1)}$

✓ **Answer: (C)** The log-likelihood is maximized at $\hat{\mu} = \bar{X}$, which is also UMVUE and unbiased.

4.3 Interval Estimation (Confidence Intervals)

Confidence Intervals

CI for μ (σ known):

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

CI for μ (σ unknown, n small):

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$$

CI for proportion p :

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Common z values: $z_{0.025} = 1.96$ (95%), $z_{0.005} = 2.576$ (99%), $z_{0.05} = 1.645$ (90%)

LaTeX Code

```
\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \quad \bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}
```

4.3.1 MCQs: Confidence Intervals

Q8. 95% CI uses $z_{\alpha/2} =$

- (A) 1.645 (C) 2.33
(B) 1.96 (D) 2.576

✓ **Answer: (B)** For 95% CI, $\alpha = 0.05$, $z_{0.025} = \boxed{1.96}$.

Q9. $n = 100$, $\bar{X} = 50$, $\sigma = 10$. 95% CI for μ :

- (A) (48.04, 51.96) (C) (47, 53)
(B) (49, 51) (D) (48, 52)

✓ **Answer: (A)** $ME = 1.96 \times \frac{10}{\sqrt{100}} = 1.96$. CI: $(50 - 1.96, 50 + 1.96) = \boxed{(48.04, 51.96)}$.

Q10. A wider CI means:

- (A) Higher confidence OR smaller sample size (C) Both A and B
(B) Lower precision (D) Neither A nor B

✓ **Answer: (C)** Wider CI $\Leftrightarrow$ higher confidence level or smaller sample size, both of which reduce precision.

5. UNIT 5: HYPOTHESIS TESTING

5.1 Foundations of Hypothesis Testing

Key Concepts

Null Hypothesis H_0 : Statement of no effect / no difference (assumed true until evidence rejects it).

Alternative Hypothesis H_1 : The claim we want to test.

Type I Error (α): Reject H_0 when H_0 is true. (α = significance level)

Type II Error (β): Fail to reject H_0 when H_0 is false.

Power: $1 - \beta$ = probability of correctly rejecting H_0 .

p -value: Probability of observing the test statistic as extreme or more extreme, assuming H_0 true. Reject H_0 if p -value $< \alpha$.

Common Test Statistics

One-sample z -test (σ known):

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

One-sample t -test (σ unknown):

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$$

Two-sample t -test (equal variances):

$$t = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{1/n_1 + 1/n_2}}, \quad S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Chi-squared test (goodness of fit / independence):

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

LaTeX Code

```
z=\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}, \quad t=\frac{\bar{X}-\mu_0}{S/\sqrt{n}}\sim t_{n-1}
\chi^2=\sum_i \frac{(O_i-E_i)^2}{E_i}
```

5.1.1 MCQs: Hypothesis Testing

Q1. Rejecting H_0 when it is actually true is:

- | | |
|----------------------|-------------------------------|
| (A) Type II Error | (C) Type I Error (α) |
| (B) Correct decision | (D) Power |

✓ **Answer: (C)** Type I Error = rejecting a true null hypothesis. Its probability = α = significance level.

Q2. p -value = 0.03 and $\alpha = 0.05$. Conclusion:

- | | |
|--------------------------|---------------------------------|
| (A) Fail to reject H_0 | (C) Accept H_1 with certainty |
| (B) Reject H_0 | (D) Need more data |

✓ **Answer: (B)** Since $p = 0.03 < \alpha = 0.05$, we **reject** H_0 at the 5% significance level.

Q3. Power of a test is:

- | | |
|--------------|------------------|
| (A) α | (C) $1 - \alpha$ |
| (B) β | (D) $1 - \beta$ |

✓ **Answer: (D)** Power = $1 - \beta = P(\text{reject } H_0 | H_0 \text{ false})$ = probability of correctly detecting a real effect.

Q4. $n = 36$, $\bar{X} = 52$, $\mu_0 = 50$, $\sigma = 12$. Test statistic $z =$

- | | |
|-------|---------|
| (A) 2 | (C) 0.5 |
| (B) 1 | (D) 3 |

✓ **Answer: (B)** $z = \frac{52 - 50}{12/\sqrt{36}} = \frac{2}{12/6} = \frac{2}{2} = \boxed{1}$.

Q5. Chi-squared test for independence in a contingency table uses degrees of freedom:

- | | |
|----------------------|-----------------|
| (A) $(r - 1)(c - 1)$ | (C) $r + c - 1$ |
| (B) rc | (D) $r + c - 2$ |

✓ **Answer: (A)** For a $r \times c$ contingency table, the χ^2 test has $(r - 1)(c - 1)$ degrees of freedom.

Q6. Increasing sample size n (all else equal):

- | | |
|----------------------------|---|
| (A) Increases Type I error | (C) Decreases Type II error (increases power) |
| (B) Decreases power | (D) Has no effect on errors |

✓ **Answer: (C)** Larger $n \Rightarrow$ smaller SE $\Rightarrow$ larger test statistic for same true effect $\Rightarrow$ greater power $\Rightarrow$ smaller Type II error β .

6. UNIT 6: REGRESSION AND CORRELATION

6.1 Simple Linear Regression

Simple Linear Regression Model

$$Y = \beta_0 + \beta_1 X + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

Y : response variable, X : predictor, β_0 : intercept, β_1 : slope, ε : error.

Least Squares Estimators

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x \quad (\text{fitted/predicted value})$$

$$e_i = y_i - \hat{y}_i \quad (\text{residual})$$

Pearson Correlation Coefficient

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$-1 \leq r \leq 1; \quad r^2 = \text{coefficient of determination}$$

$$r^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}$$

ANOVA Decomposition for Regression

$$\underbrace{\sum (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{\text{SSR}} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{\text{SSE}}$$

$$s^2 = \frac{\text{SSE}}{n - 2} \quad (\text{estimate of } \sigma^2)$$

LaTeX Code

```
\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2},
\quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}
```

6.1.1 MCQs: Regression and Correlation

Q1. $\sum x_i = 30$, $\sum y_i = 50$, $n = 5$. Then $\bar{x}$ and $\bar{y}$:

(A) 6 and 10

(C) 5 and 10

(B) 10 and 6

(D) 6 and 8

✓ **Answer: (A)** $\bar{x} = 30/5 = 6, \bar{y} = 50/5 = 10$.

Q2. $S_{xx} = 40, S_{xy} = 20$. $\hat{\beta}_1 =$

- (A) 2 (C) 0.8
(B) 0.5 (D) 20

✓ **Answer: (B)** $\hat{\beta}_1 = S_{xy}/S_{xx} = 20/40 = \boxed{0.5}$.

Q3. $r = 0.9$. Coefficient of determination $r^2 =$

- (A) 0.81 (C) 0.45
(B) 0.9 (D) 0.3

✓ **Answer: (A)** $r^2 = 0.9^2 = \boxed{0.81}$. This means 81% of variation in Y is explained by X .

Q4. If $r = -1$, the regression line:

- (A) Is horizontal (C) Does not exist
(B) Has a perfect negative linear relationship (D) Has slope +1

✓ **Answer: (B)** $r = -1$ means perfect negative linear correlation: all points lie exactly on a line with negative slope.

Q5. Regression line always passes through:

- (A) Origin $(0, 0)$ (C) $(\bar{x}, \bar{y})$
(B) $(\bar{x}, 0)$ (D) $(0, \bar{y})$

✓ **Answer: (C)** The least squares regression line always passes through $(\bar{x}, \bar{y})$, the centroid of the data.

Q6. $SST = 100, SSE = 40$. Then $r^2 =$

- (A) 0.4 (C) 0.8
(B) 0.6 (D) 2.5

✓ **Answer: (B)** $r^2 = 1 - \text{SSE}/\text{SST} = 1 - 40/100 = \boxed{0.6}$.

7. COMPLETE LATEX FORMULA REFERENCE

Probability Fundamentals

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(A^c) = 1 - P(A)$$

$$P(B_k|A) = \frac{P(A|B_k)P(B_k)}{\sum_i P(A|B_i)P(B_i)} \quad (\text{Bayes})$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}, \quad {}^nP_r = \frac{n!}{(n-r)!}$$

Discrete Distributions Summary

Distribution	PMF $P(X = k)$	$E[X]$	$\text{Var}(X)$
$B(n, p)$	$\binom{n}{k} p^k q^{n-k}$	np	npq
$\text{Pois}(\lambda)$	$\frac{e^{-\lambda} \lambda^k}{k!}$	λ	λ
$\text{Geom}(p)$	$(1-p)^{k-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
$\text{HGeom}(N, K, n)$	$\frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$	$\frac{nK}{N}$	$n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}$
$\text{NegBin}(r, p)$	$\binom{k-1}{r-1} p^r (1-p)^{k-r}$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$

Continuous Distributions Summary

Distribution	PDF $f(x)$	Support	$E[X]$	$\text{Var}(X)$
$U(a, b)$	$\frac{1}{b-a}$	$[a, b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$N(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$\mathbb{R}$	μ	σ^2
$\Gamma(\alpha, \beta)$	$\frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha}$	$x > 0$	$\alpha\beta$	$\alpha\beta^2$
$\text{Beta}(\alpha, \beta)$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$	$(0, 1)$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

All Key LaTeX Codes

```
% Normal PDF:
f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}

% Bayes:
P(B_k|A)=\frac{P(A|B_k)P(B_k)}{\sum_i P(A|B_i)P(B_i)}

% Variance decomposition:
\mathrm{Var}(X)=E[X^2]-(E[X])^2

% Regression slope:
\hat{\beta}_1=\frac{S_{xy}}{S_{xx}}=\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}

% Correlation:
```

```
r=\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}},\quad -1\leq r\leq 1
```

% z-test:

```
z=\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}},\quad t=\frac{\bar{X}-\mu_0}{S/\sqrt{n}}\sim t_{n-1}
```

% CI for mean:

```
\bar{X}\pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}
```

8. MASTER QUICK REVISION TABLE

Topic	Key Formula	PSC Exam Tip
Addition Rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Set $P(A \cap B) = 0$ if mutually exclusive
Bayes' Theorem	$P(B A) = \frac{P(A B)P(B)}{P(A)}$	Always draw a probability tree first
Binomial	$E = np, \text{Var} = npq$	Identify n, p from the problem context
Poisson	$E = \text{Var} = \lambda$	Approximates Binomial when $n > 50, p < 0.1$
Normal Std.	$Z = (X - \mu)/\sigma$	$P(-1.96 < Z < 1.96) = 0.95$ (memorise!)
Exponential	$E[X] = 1/\lambda$, memoryless	Only continuous memoryless distribution
Variance Formula	$\text{Var} = E[X^2] - (E[X])^2$	Use this shortcut rather than definition
Regression Slope	$\hat{\beta}_1 = S_{xy}/S_{xx}$	r^2 = proportion of variance explained
CI (known σ)	$\bar{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}$ (95%)	Wider CI $\Leftrightarrow$ higher confidence
Hypothesis Test	Reject if $ z > z_{\alpha/2}$ or $p < \alpha$	Power = $1 - \beta$; larger n increases power
Covariance	$\text{Cov} = E[XY] - E[X]E[Y]$	$\rho = 0$ means uncorrelated, not necessarily independent
Chi-squared	$\chi^2 = \sum (O - E)^2 / E$	df = $(r - 1)(c - 1)$ for independence test

Table 1: Master Quick Revision: Probability & Statistics for AJKPSC BPS-17

9. 4-WEEK STUDY PLAN

Week	Topics	Daily Goals	Hours/Day
1	Sample space, Events, Axioms, Counting, Conditional Prob., Bayes' Theorem	Solve 20 Bayes problems; master probability trees and contingency tables	3–4 hrs
2	Random Variables, Discrete Distributions (Binomial, Poisson, Geometric, Hypergeometric)	Memorise all PMFs, E , Var; practice 25 distribution problems per day	4 hrs
3	Continuous Distributions, Normal, Exponential, Gamma; MGF; Joint Distributions	Master z -score tables; practice Normal calculations; derive MGFs	4 hrs
4	Estimation (Point & Interval), Hypothesis Testing, Regression, Mock Tests	Solve complete past AJKPSC papers; focus on t -test, χ^2 , regression	4–5 hrs

Table 2: 4-Week AJKPSC BPS-17 Probability & Statistics Study Schedule

Important PSC Exam Tips for Probability & Statistics

- Normal distribution:** Memorise $z_{0.025} = 1.96$ (95%), $z_{0.005} = 2.576$ (99%), $\Phi(1.96) \approx 0.975$.
- Bayes' theorem:** Always identify the partition, priors, and likelihoods before computing.
- Poisson vs Binomial:** Poisson when $n > 50$, $p < 0.1$, rate $\lambda = np$; both have different variance formulas.
- Memoryless:** Only **Geometric** (discrete) and **Exponential** (continuous) are memoryless.
- Unbiased estimators:** $\bar{X}$ for μ ; $S^2 = \sum (X_i - \bar{X})^2 / (n - 1)$ for σ^2 .
- Hypothesis testing:** $p\text{-value} < \alpha \Rightarrow \text{reject } H_0$; Type I (α) vs Type II (β) errors.
- Regression:** r^2 = proportion of variance explained; line passes through $(\bar{x}, \bar{y})$.
- Covariance zero:** $\nRightarrow$ independence in general; independence $\Rightarrow$ covariance zero.
- Chi-squared test:** $\text{df} = (r - 1)(c - 1)$ for independence; $E_{ij} = R_i C_j / n$ for expected frequencies.

10. **PSC pattern:** Binomial, Poisson, Normal, Bayes, regression, and CI questions appear in *every* AJKPSC/FPSC paper.

Exam Practice Resource

For further exam practice, the following resource provides a comprehensive collection of solved MCQs:
Access the material here

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Best of luck in your examination!